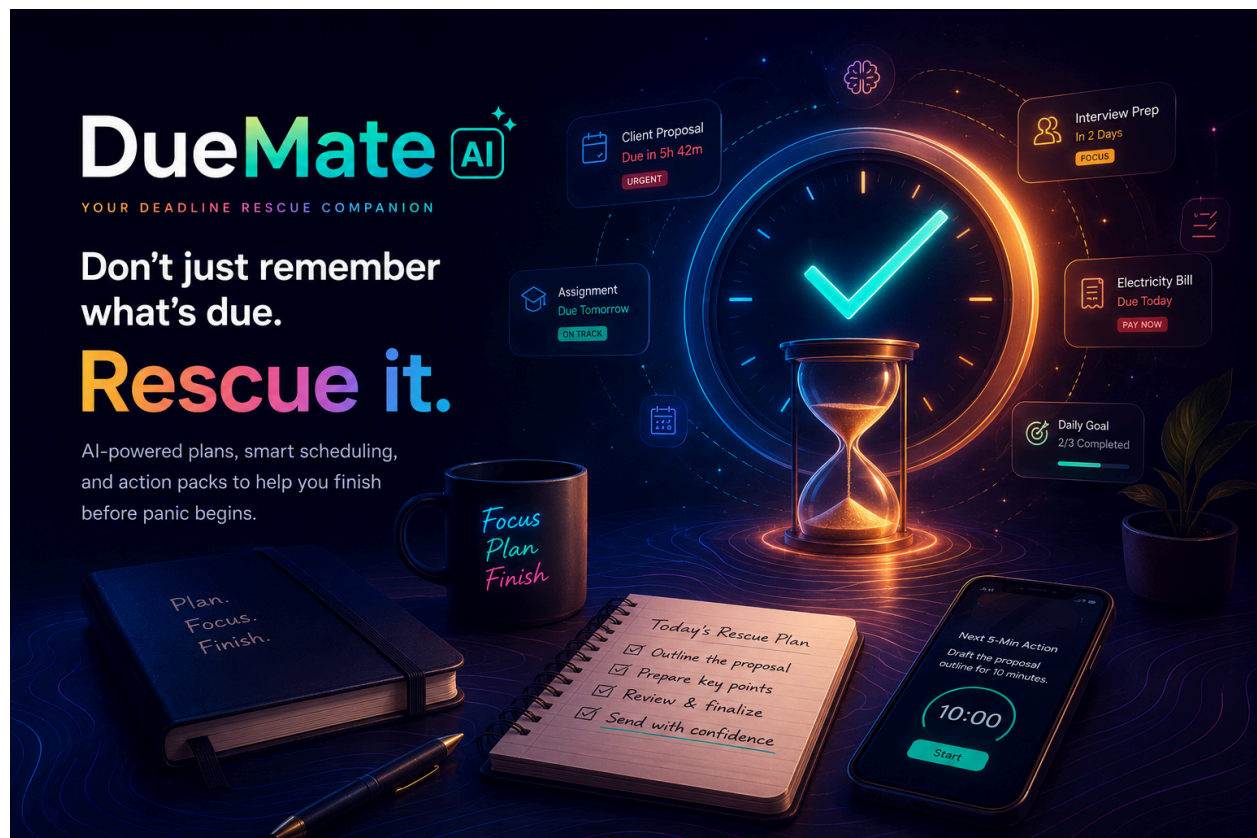


DueMate AI — Project Description



Project Links

Deployed Application Link:

<https://duemate-ai-920403302790.asia-southeast1.run.app>

GitHub Repository Link:

<https://github.com/Jidnyasa-P/DueMate>

Problem Statement Selected

Problem Statement 1 — The Last-Minute Life Saver

Students, professionals, entrepreneurs, freelancers, and job seekers often miss deadlines, assignments, meetings, bill payments, interviews, and important commitments. Traditional productivity tools mainly provide passive reminders, but reminders alone do not help users actually begin, prioritize, plan, or complete their work.

DueMate AI addresses this problem by acting as a proactive deadline rescue companion. Instead of only reminding users what is due, it helps them understand what is risky, what to do first, how to break work into manageable actions, and how to recover when a deadline is already close.

Solution Overview

DueMate AI is a premium role-adaptive productivity companion designed to help users finish what is due before panic begins.

The product transforms stressful deadlines into structured rescue plans. A user can register, select their role, complete a short onboarding flow, and receive a workspace that adapts to their deadline type, work style, reminder preference, and energy pattern.

The core idea is simple:

DueMate AI does not just tell users that a task is due. It gives them the safest next move to complete it.

When a user adds or selects a task, DueMate AI generates a practical rescue plan containing:

- the safest next action,
- a first 5-minute starting step,
- micro-steps,
- time blocks,
- risk factors,
- a fallback plan,
- role-specific action assets,
- and a delay simulation showing what happens if the user postpones the task.

The app is designed to support users during real deadline pressure by reducing overwhelm, breaking large work into smaller actions, and helping them regain control through guided execution.

Key Features

1. Role-Adaptive Workspace

DueMate AI personalizes the experience for different users:

- **Students** — assignments, exams, study plans, lab reports, revision habits.
- **Professionals** — meetings, reports, presentations, payments, work deadlines.

- **Entrepreneurs** — client delivery, fundraising, invoices, launch plans, pitch preparation.
- **Freelancers** — client tasks, revisions, proposals, invoices, portfolio work.
- **Job Seekers** — interviews, applications, resume improvement, follow-ups, skill practice.

The app does not simply change labels; it changes the planning logic, sample tasks, habits, action packs, and recommended next steps based on the selected role.

2. Rescue Plan Engine

The Rescue Plan Engine converts a task into a clear completion path. It generates:

- task understanding,
- risk summary,
- safest next move,
- first 5-minute action,
- micro-steps,
- time blocks,
- fallback plan,
- definition of done,
- and role-specific action packs.

This directly moves the app beyond passive reminders by helping the user take meaningful action.

3. Delay Simulator

The Delay Simulator helps users understand the real cost of postponing work. It estimates:

- increased deadline risk,
- stress increase,
- overlapping conflicts,
- recovery time,
- what to skip,
- what to do first,
- and an honest recommendation.

This makes procrastination visible and helps users make better decisions before it is too late.

4. Panic Button Emergency Autopilot

The Panic Button is designed for high-pressure deadline situations. It helps users reduce the task scope when the original ideal plan is no longer realistic.

It can generate:

- a minimum viable task scope,
- what can be safely cut,
- a basic passing-quality target,
- extension or update email drafts,
- and a low-resistance writing sandbox.

This feature is especially useful when the user is overwhelmed and needs a practical recovery path instead of another reminder.

5. Bio-Rhythm and Fatigue-Aware Scheduler

DueMate AI considers the user's energy state while recommending what to do next. The user can choose states such as high focus, creative hyperfocus, post-lunch slump, or brain fog.

Based on the selected state, the app can reorder tasks and suggest more realistic work actions. For example, complex analytical work can be prioritized during high-energy periods, while low-friction admin tasks can be suggested during low-energy periods.

6. Body Doubling and Interactive Focus Coach

The app includes a simulated co-working experience where users can work alongside a focus companion. This feature supports users who struggle with starting work alone.

It includes:

- selectable focus peers,
- synchronized focus timers,
- progress updates,
- supportive prompts,
- and a whisper box for writing down starting anxiety.

This helps users move from task avoidance to active progress.

7. Procrastination Style Analytics

DueMate AI includes a self-awareness dashboard that helps users understand their delay patterns. It identifies procrastination tendencies such as perfectionism, avoidance, underestimation, and panic-driven work.

The goal is not only to complete one task, but to help users recognize why they repeatedly miss deadlines and improve their habits over time.

8. Multi-Agent Cognitive Tracer

The app uses a multi-agent planning structure to show how the rescue plan is created. The agent flow includes:

- Capture Agent,
- Risk Agent,
- Planner Agent,
- Scheduler Agent,
- Execution Agent,
- Coach Agent,
- Reflection Agent.

The trace is shown in a collapsible format so the dashboard remains clean while still demonstrating transparent AI reasoning and planning depth.

9. Voice Input Support

DueMate AI uses browser speech recognition to let users capture deadlines by speaking. This is useful when users remember an urgent task suddenly and want to record it quickly without filling out a long form.

10. Appearance and Focus Modes

The app supports multiple workspace modes:

- Light mode,
- Dark mode,
- Read mode,
- Focus mode,
- Panic visual state,
- and role-adaptive styling.

This makes the product more accessible and adaptable to different user preferences and stress levels.

11. Local Persistence and Offline-Friendly Fallbacks

DueMate AI stores user profile data, tasks, habits, and workspace settings locally in the browser. If Gemini is unavailable or rate-limited, the app still generates structured rescue plans using local fallback logic.

This ensures the core product remains usable even when AI services are temporarily unavailable.

Technologies Used

Frontend

- React
- TypeScript
- Vite
- Tailwind CSS
- Lucide React icons
- Motion-based interface animations
- Responsive single-page application architecture

Backend

- Node.js
- Express
- Server-side Gemini API route handling
- JSON-based API endpoints
- Local fallback planning engine

Data Persistence

- Browser localStorage for demo profile, tasks, habits, preferences, and tour completion
- Local task and habit persistence per user profile
- No paid database required for the current version

Browser APIs

- Web Speech API for voice-based deadline capture
- Browser Notification API for optional local reminders
- Clipboard API for copying generated drafts or links

Productivity Features

- Rescue plan generation
 - Delay simulation
 - Focus timer
 - Role-specific task templates
 - Action packs
 - Calendar-style internal planning
 - Habit tracking
 - Tour/onboarding flow
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Google Technologies Utilized

Google AI Studio

DueMate AI was built using Google AI Studio's app development workflow. The project structure follows the AI Studio-generated repository format and is designed as a full-stack web application.

Gemini API

The app integrates Gemini through the `@google/genai` SDK on the server side. Gemini is used to generate structured deadline rescue plans, role-specific action packs, multi-agent traces, and delay simulations.

Gemini-generated outputs are structured as JSON to make the responses reliable and directly usable inside the app interface.

Google Cloud Deployment

The application is deployed publicly on a Google Cloud `run.app` domain, making it accessible for evaluation and real usage.

Google Calendar-Compatible Planning

The app is designed around calendar-based productivity workflows and supports calendar-style planning. The roadmap also includes deeper Google Calendar integration for external synchronization.

Why DueMate AI Is Different

Most productivity tools focus on organizing tasks. DueMate AI focuses on rescuing tasks.

Instead of only showing a list of deadlines, it answers the question users actually have under pressure:

“What should I do right now so I still have a chance to finish?”

The product combines role-adaptive planning, delay simulation, panic recovery, focus coaching, and habit reflection into one practical deadline rescue experience.

Its main differentiator is that it helps users move from anxiety to action through small, realistic, and personalized next steps.

Current Implementation Status

The current version includes:

- public deployed application,
 - public GitHub repository,
 - landing page,
 - login/register flow,
 - role onboarding,
 - role-adaptive sample tasks,
 - dashboard,
 - tasks and habits,
 - rescue plan generation,
 - Gemini API integration,
 - local fallback planning,
 - delay simulator,
 - panic rescue suite,
 - focus coach,
 - procrastination analytics,
 - appearance modes,
 - local persistence,
 - and responsive UI.
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Future Scope

Future improvements can include:

- secure OAuth-based Google Calendar integration,
- cloud sync across devices,
- reminder scheduling across mobile and desktop,
- deeper file analysis for assignment or project requirements,

- collaborative deadline rescue rooms,
 - and personalized long-term productivity reports.
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Conclusion

DueMate AI is a practical AI-powered deadline rescue companion that helps users plan, prioritize, and complete urgent commitments before deadlines are missed. By focusing on action instead of passive reminders, it provides a more useful and human-centered approach to productivity for students, professionals, entrepreneurs, freelancers, and job seekers.